

Milestone 4 – Final Review and Delivery

Project: Unified Military Analytics and Comparison Dashboard

Platform: Power BI

Internship: Infosys Springboard Virtual Internship 6.0

Module 7 – Testing, Validation & Debugging Report

1. Objective

The objective of Module 7 was to perform comprehensive validation of all filters, DAX measures, KPI outputs, aggregation logic, navigation controls, and visual accuracy. This ensured the final Power BI dashboard is stable, accurate, and presentation-ready.

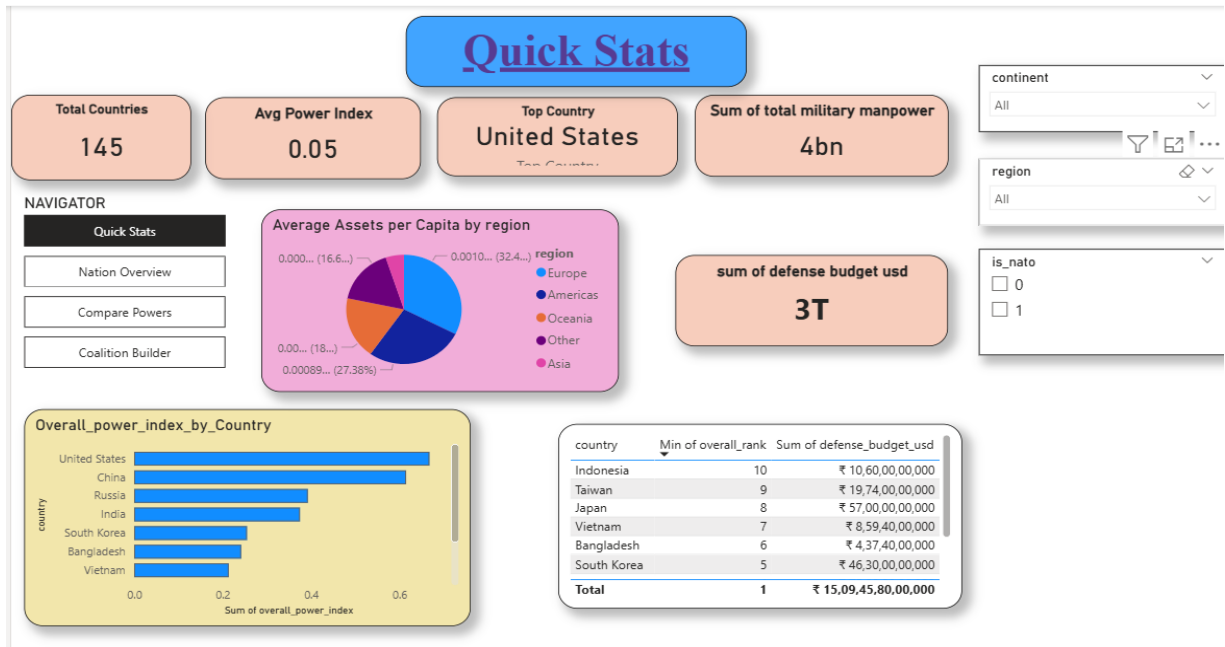
2. Functional Testing

- Tested all country, region, continent, and NATO slicers.
- Verified multi-country selection logic for comparison and coalition analysis.
- Confirmed navigation buttons redirect correctly across pages.
- Validated dynamic updating of KPIs and visuals.

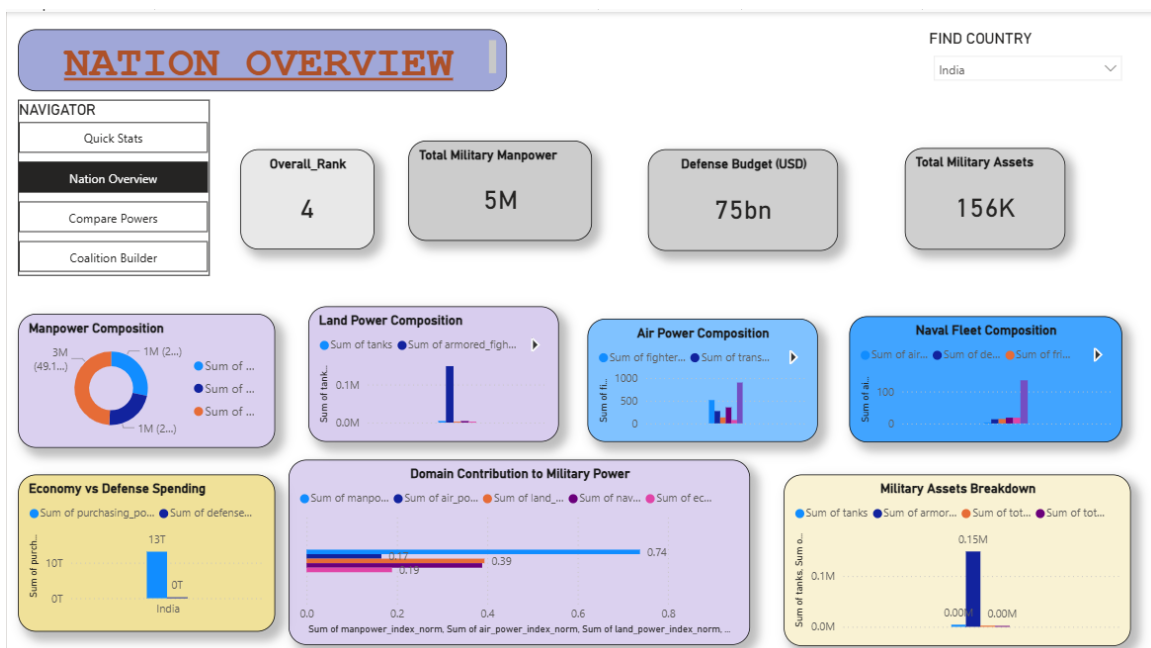
3. Manual Data Validation (Spot Checks)

Dedicated validation pages were created to manually verify totals for manpower, assets, defense budget, domain contribution, and derived measures such as Budget per Soldier and Efficiency Index.

QUICK STATS:



Nation Overview:



Nation Overview:



4. Calculation Verification

- Budget per Soldier verified against total budget and manpower.
- Coalition totals verified using manual aggregation tables.
- Domain contribution percentages cross-checked.
- No division-by-zero or aggregation errors detected.

5. UI & Layout Debugging

- Fixed alignment and spacing inconsistencies.
- Standardized card formatting and color themes.
- Corrected tooltip labeling issues.
- Ensured consistent naming conventions across visuals.

6. Testing Outcome

All modules passed validation checks. No functional bugs were identified. The dashboard is verified end-to-end and ready for final submission.

Module 8 – Documentation & GitHub Release

1. Objective

Module 8 focused on finalizing documentation, organizing the repository structure, preparing the Power BI dashboard for submission, and ensuring professional GitHub presentation.

2. Final Deliverables

- Integrated Power BI Dashboard: Global_Military_Powers_Stats.pbix
- Complete GitHub repository with structured milestone folders
- README with scraping, KPI, and dashboard usage details
- Validation and QA documentation

3. Repository Organization

The final repository includes:

- Milestone 1 – Data Collection & Cleaning
- Milestone 2 – KPI Engineering
- Milestone 3 – Dashboard Development
- Milestone 4 – Testing & Documentation

4. Dashboard Usage Guide

1. Open Power BI Desktop.
2. Load Global_Military_Powers_Stats.pbix.
3. Use slicers to filter by country, region, or alliance.
4. Navigate between pages using dashboard buttons.
5. Analyze comparison and coalition insights.

5. Project Conclusion

The Unified Military Analytics and Comparison Dashboard demonstrates an end-to-end data pipeline including scraping, cleaning, KPI engineering, dashboard development, validation, and documentation. The project is fully integrated, tested, documented, and ready for academic evaluation and portfolio presentation.